

Project Proposal

infrared.city-1: Tree detection

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1 Project Description

1.1 Motivation and Project Overview

As urban areas are expanding and global temperatures rise due to climate change, being able to understand how trees can help cool cities has become increasingly important [1]. Furthermore, trees reduce the risk of flooding in cities and promote health and wellbeing [2]. Our project aims at taking the first step in the direction of better tree coverage in urban areas by creating a model which is able to predict tree locations using satellite imagery of densely populated areas.

1.2 Data Sources

For our project we use publicly available Sentinel-2 satellite images with 10-meter spatial resolution and four color channels, namely red, green, blue and near infrared. The images are retrieved via the API of the Copernicus Programme and are provided by the European Space Agency [3]. As seasonality has a great impact on the appearance of trees, imagery of different seasons will be used to train the model. Additionally, we will use the publicly available data set on every tree in the city of Vienna “Baumkataster” to train our model on exact tree locations [4].

1.3 Model

The model used to predict the tree locations is the U-Net deep learning model which is designed for image segmentation. It is a special type of Convolutional Neural Network and consists of a contracting path and an expanding path. In the contracting path convolution and pooling layers are used to extract features from the original imagery – which in this project will consist of four pictures for each season and four color channels each as described above. The expanding path consists of the same structure of convolution and pooling layers but used in the other direction to up-sample the before contracted imagery [5]. As suggested in similar work this project will use skip connections between the expanding and contracting path as well as a temporal attention algorithm to prioritize more

meaningful images [6]. The Binary Cross-Entropy loss between the predicted locations and the actual locations from the “Baumkataster” will be used to train the model.

1.4 Project Scope

Further analysis on the impact of trees on, for example, heat regulation in cities and flooding prevention are out of the scope of this project and could be the topic of future research.

2 BEAM

The figure below presents the BEAM representation of the proposed tree detection system, illustrating the data sources, training and inference processes, model architecture, evaluation pipeline, and associated risks.

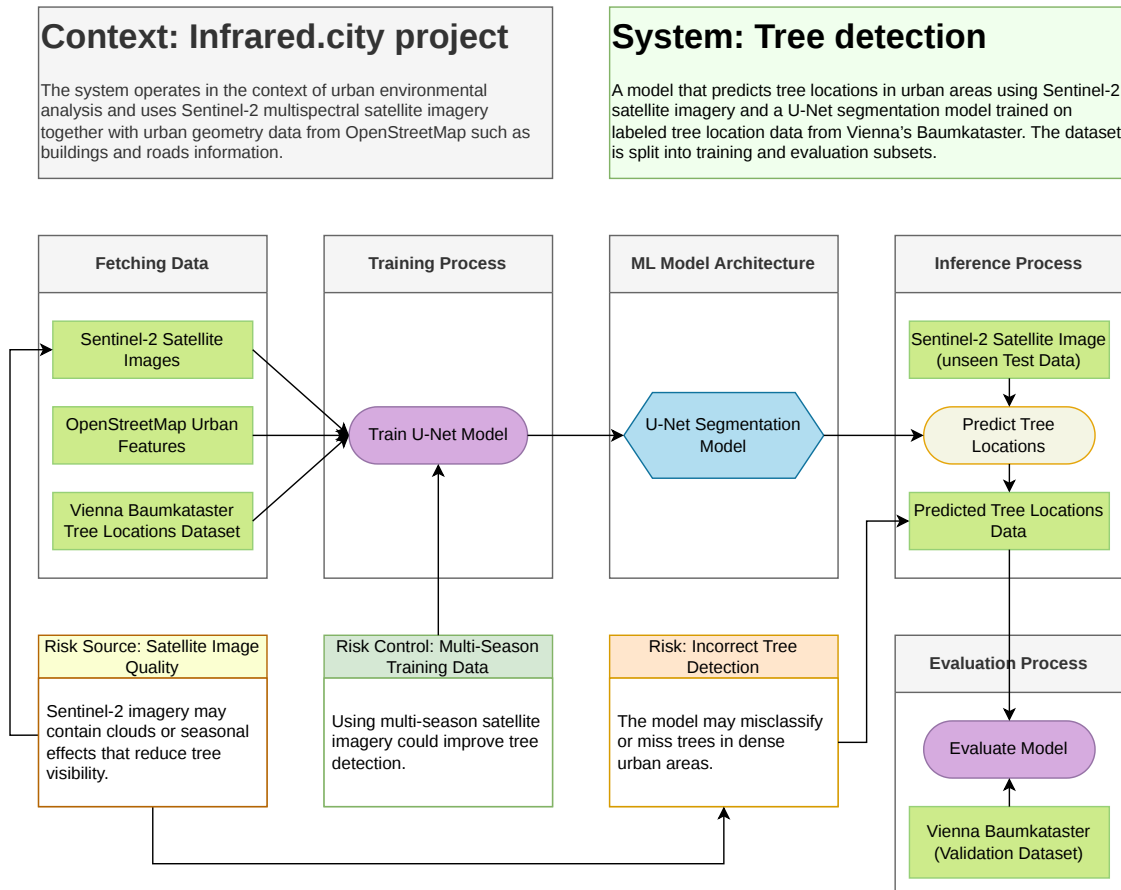


Figure 1: BEAM model of the proposed tree detection system.

3 Timeplan

TODO set of Milestones and Tasks, including deliverables

4 Communication outline

This section describes the communication stiles we are committed to use over the project.

4.1 Written communication

We will use two Teams in Microsoft Teams for our basic communication.

1. **DsLab26S_infrared.city_1**: This Team contains two channels for two different approaches. The **General** Channel is used as communication level with Kavita Surana (our connection to the Lecture and the WU). The other channel (**team intern**) is only accessable by the team itself and is our main chat for communication over the project.

2. **DsLab_infrared.city_1-coaches**: This Team is the basic for communication with the data coaches from infrared-city. We were asked to separate this from the main communication source, but still keep an Microsoft Team Teams for this. So we created an different Team

4.2 Regular Meetings

Our Meetings will be hold via Microsoft Teams in most cases. Possible that we are able to meet in person for the one or other defined meeting. But regulary we plan to keep the communication remote or at least hibrate.

We organized already two regulary weekly meetings for basic communication.

1. **Regular Team Meeting**: On Thursdays 19:00 we will meet as a team remote and talk about how to continue the work over the next week.

2. **Meeting with Data Coaches from infrared.city** On Fridays 12:30 we scheduled a short regulary meeting with the data coaches from infrared.city.

3. **Irregular Meetings with Kavita Surana**: In cases we need feedback for something from the WU site we schedule a meeting with her. To easily schedule the meeting we got an booking site for her time.

4.3 Documentation of Meetings

Because of the many meetings it is possible that the one or other person is not able to partizipate every time. To still be able to communicate with absent people we want to write meeting notes and documentation of the meeting output after every meeting.

4.4 GIT

Every written result (documenation and project related) will be added to an already created GitHub repository. The repository is accessable by the team. See https://github.com/nikmaster/wu_dslab_infrared-city.

5 Team Roles

The main project team consist of 4 persons which are responsible for different focus points. But still commit to the project and in case one person will be stucked the focus has to be to work together and find a way to tackle the issue.

Person	Role	Tasks
Dominik Oberhumer	Organisation of Team	Preparation of Tools, Reminding of Deliverables and Timeplan
Moritz Hörmansdorfer	Data Scientist	Machine Learning Training of Model
Nina Salnikow	Data Engineer	Fetching and Preparation of Data
Sara Klasová	Data Scientist	EDA (Exploratory Data Analysis)

Besides of this persons we also have additional support from people which are related to the project:

1. **Kavita Surana**: is our link to the lecture at the WU. And will help us regarding infrastructure and gives the needed points for grading at the university

2. **Oana Taut** and **Vasiliki Fragkia** are our data coaches from infrared.city. They will support us with different approaches and sources how we can use the data to archive our goal.

6 Literature

Following additional papers are planned to be used as sources:

1. A review of individual tree crown detection and delineation from optical remote sensing images [7]
2. Integrating Traditional and Deep Learning Methods to Detect Tree Crowns in Satellite Images [8]
3. A Sentinel-2 machine learning dataset for tree species classification in Germany [9]
4. Object-based classification of urban plant species from very high-resolution satellite imagery [10]
5. Improving Urban Tree Species Classification with High Resolution Satellite Imagery and Machine Learning [11]

References

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- [5] A. I. Aramendia, “Decoding the u-net: A complete guide,” 2024. Abgerufen: 2026-03-15.
- [6] V. S. F. Garnot and L. Landrieu, “Panoptic segmentation of satellite image time series with convolutional temporal attention networks,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 4872–4881, 2021.

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